

Total No. of Questions : 8]

SEAT No. :

PC1726

[6353]-43

[Total No. of Pages : 2

T.E. (Computer Engineering)

DATA SCIENCE AND BIG DATA ANALYTICS

(2019 Pattern) (Semester-II) (310251)

Time : 2 ½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right indicate full marks
- 3) Neat Diagram must be drawn wherever necessary.
- 4) Assume suitable data if necessary.

- Q1)** a) Draw data analytics life cycle diagram and briefly explain its phases. [6]
- b) Explain the various key roles for a successful analytics project. [6]
- c) What are various sources of Big data. [6]

OR

- Q2)** a) Describe few applications of Big Data Analytics. [6]
- b) Explain Data preparation phase of data analytics lifecycle. [6]
- c) List common tools used for model building phase of data analytic. [6]

- Q3)** a) Define and explain Entropy and Information gain. Calculate the entropy of the following distribution [9]

Fruit Color	Taste	Count
Yellow	Sweet	10
Red	Sweet	5
Green	sour	15
Orange	sour	5

- b) Explain Naïve Bayes Classifier. [8]

OR

P.T.O.

- Q4) a)** Explain Apriori algorithm with suitable example. [9]
b) Describe different categories of analytics [8]

- Q5) a)** What is Hierarchical clustering? Explain hierarchical clustering algorithms. [9]
b) Write a note on: [8]
i) Holdout method
ii) k-Fold Cross-Validation

OR

- Q6) a)** What is Text analysis? Explain the different steps involved in the text analysis. [9]
b) Write a note on Social network analysis. What are the applications of Social network analysis? [8]

- Q7) a)** Explain Hive architecture with suitable diagram. Describe characteristics and features of hive. [9]
b) Describe the data visualization tool Tableau. [9]

OR

- Q8) a)** What is data visualization and objectives of data visualization? Why it is difficult visualize Big Data? [9]
b) Write a note on Microsoft Power BI and Qlik [9]

